Bob’s Pizza Emporium

Software Development Plan
10/21/25
Joshua Swanson, Kelly Byrne, Alison Michailof, Noah Allen, Caelyn Ruiz
Version 2.0

The Quintessentials

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Revision History

Date	Version	Description	Author(s)
9/18/2025	1.0	Initial Version	Joshua Swanson, Kelly Byrne, Alison Michailof, Noah Allen, Caelyn Ruiz

10/21/2025	2.0	Second Version	Joshua Swanson, Kelly Byrne, Alison Michailof, Noah Allen, Caelyn Ruiz

1. Introduction

1.1 Purpose

The purpose of this document is to provide all the details necessary to carry out the project to completion. The document will be used by the customer, project manager, and the team to ensure the plan is followed to completion within the determined time frame.

1.2 Scope

This Software Development Plan outlines the work implementations that will be used by this project, including the project's activation. Information regarding individual iterations will be available in section 4.2.2. All work outlined in this document is based on product requirements as they were interpreted at the time of writing and will be fully detailed in the Software Requirements Specification (SRS).

1.3 Definitions, Acronyms, and Abbreviations

Acronym	Meaning
SDD	Software Design Description
SDP	Software Development Plan
SRS	Software Requirements Specification
STP	Software Test Plan

1.4 Document References

Document Title	**Version**	**Date**	**Author(s)**
Software Requirements Specification	1.0	9/18/2025	Joshua Swanson, Kelly Byrne, Alison Michailof, Noah Allen, Caelyn Ruiz

1.5 System Overview

The software being developed will be a system for employees of a Pizza Shop to record orders from patrons, in addition to providing a receipt of sale. The prices for all items should be configurable, as well as the tax that is applied to the order total.

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

The objective of this project is to create a configurable application for a pizza restaurant. A server employee will use this application to take orders.

2.2 Assumptions and Constraints

All team members will be available for the full duration of the project and complete assigned tasks on schedule. All development will be completed using the agreed-upon tools, without the need for additional resources. Requirements defined in the SRS will remain stable throughout the project unless formally revised. Development will follow the waterfall model, meaning progress will depend upon completion of prior stages.

2.3 Project Deliverables

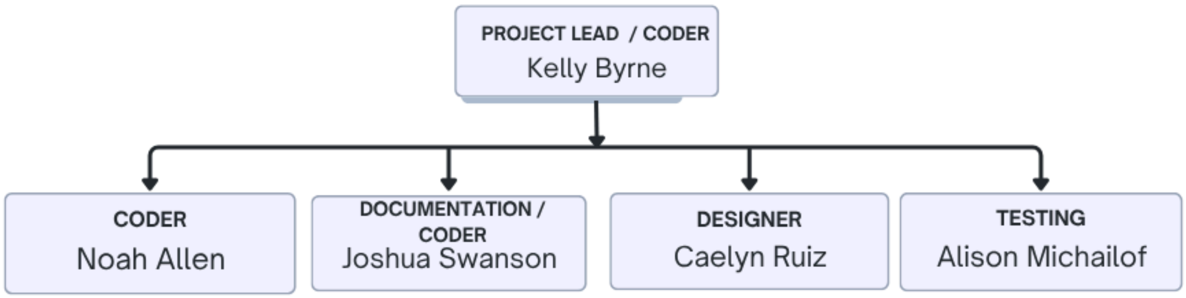
In addition to the application itself, the project will create several documents. Each document will have an original and a revised version. We will create an SDP, SRS, SDD, STP, and a Traceability Matrix. The target delivery dates for the final versions of each document, respectively, are as follows: 10/21/25, 10/28/25, 11/6/25, 11/25/25, and 10/30/25.

2.4 Evolution of the Software Development Plan

Expected Release Date	Version Number	Remarks
9/18/25	1.0	Initial version of the Software Development Plan created and submitted for review.
10/21/25	2.0	Revised version that reflects updates on Requirements Management, Quality Control, and Reporting.

3. Project Organization

3.1 Organizational Structure



3.2 External Interfaces

The only external interface the team works with is the customer, with whom the team is in contact throughout the development process to ensure everything is being done to the customer's expectations.

The customer's contact information is as follows:
Bob Tracy
301-301-7503 (Cell)
rttracy@smcm.edu

3.3 Roles and Responsibilities

Person	Project Role(s)
Kelly Byrne	Project Manager, Coder
Noah Allen	Coder
Joshua Swanson	Documentation, Coder
Caelyn Ruiz	Designer
Alison Michailof	Testing

4. Management Process

4.1 Project Estimates

Total Estimate: $0
The project will not have any expenses.

4.2 Project Plan

4.2.1 Phase Plan

Project phases will be sequential, as each relies upon the previous stage in order for the present stage to occur.

The initial stage will be Communication, which involves discussing the project with the customer and inquiring about the necessary requirements and the expectations for the project.

The second stage is planning, which involves taking the requirements from the customer and creating a development plan, as seen in this document.

The third stage is design, which involves establishing the framework for the project. This stage involves considering the requirements of the project as well as the functionality and aesthetics. It lays the groundwork for the stages to follow.

Coding is the fourth stage and will likely be the longest stage, as within this stage, the coders will build the framework and make it functional.

After coding is completed, the software will proceed into testing, where the code will be tested to ensure there are no bugs, glitches, performance issues, and that it matches the requirements outlined by the customer. If it does not, it will be returned to the coding stage.

Once the software has been thoroughly tested, it is ready for deployment. At this stage, no further development is required and only involves delivering the final, fine-tuned product to the customer.

4.2.2 Iteration Objectives

The project will only have one iteration. The objectives are as follows:

- Develop a development plan document
- Develop a requirement specification document
- Design the framework and the system
- Code the software

- Test and debug the software
- Deployment

4.2.3 Releases

The project will have a singular release, which will be the deployment of the final product.

4.2.4 Project Schedule

			week of														
Task Name	Start Date	End Date	9/1	9/8	9/15	9/22	9/29	10/6	10/13	10/20	10/27	11/3	11/10	11/17	11/24	12/1	12/8
Gather requirements	9/4	9/9															
Create SDP v1.0	9/11	9/18															
Create SDP v1.1	10/14	10/21															
Create SRS v1.0	9/18	10/2															
Create SRS v1.1	10/21	10/28															
Create SDD v1.0	9/25	10/16															
Create SDD v1.1	10/28	11/6															
Create STP v1.0	11/4	11/13															
Create STP v1.1	11/18	11/25															
Traceability Matrix v1.0	9/30	10/9															
Traceability Matrix v1.1	10/14	10/30															
Coding	9/11	12/9															
Testing	9/16	12/9															

4.2.5 Project Resourcing

This project team will consist of five members. No special training is required or will occur.

4.3 Project Monitoring and Control

The team will conduct weekly meetings to provide updates on the assigned project and check the Gantt chart for upcoming deadlines. The project manager will communicate with all team members to ensure steady progress.

To keep the separation of work, each team member will be given a task specifically to their project role that they are solely responsible for. Team members can request help from other members if progress is stagnant or if too many problems arise in their project field.

4.3.1 Requirements Management

At the beginning of each stage, the team will meet to assign tasks to each member based on their project role or any known bugs, ensuring project goals are met. All changes will be submitted as change requests, and the project manager will review them to assess their impacts on scope, schedule, and resources. Once approved, changes will be documented, incorporated into the project, and reflected in updated versions of all relevant documents.

4.3.2 Quality Control

To ensure a high-quality product, the development will follow the waterfall model, with repeated stages of testing and planning. This will start with identifying objectives, followed by risk analysis, product development, and finally evaluation. These four processes will occur frequently throughout the development cycle. Each development cycle will aim to remove any defects from the previous cycle. All identified defects will be assigned to team members for correction. The project manager will review all defects to ensure timely corrections and prevent recurrence. Each development cycle will address and remove defects found in previous stages, with testing to ensure fixes do not introduce new issues.

4.3.3 Reporting and Measurement

The project will undergo infrequent reporting cycles throughout the development span and will be done throughout the evaluation period. The success of the project will be measured through a review with the client at the end of construction. This will be done to gauge customer satisfaction, investigate problems, and highlight success. These metrics will be gathered through a survey with a variety of questions, such as “On a scale of 1-10, how easy is the product to use?” This will give the development team a better understanding of project goals and whether they were met. This will be collected and maintained in a separate document after the initial client review.

4.3.4 Risk Management

Risks will be highlighted throughout the risk analysis portion of the development cycle. Project risk will be accentuated by a member of the team. Any risks that are discovered will be noted and stored in an additional document. Individual risks will be given steps to mitigate and ultimately remove any from the project. As the project becomes more complex, so does the potential for risk. This will be a priority.

4.3.5 Configuration Management

We will use the most current versions of software or language models available during development. Programmers will be responsible for maintaining the project's source code in an organized manner in a GitHub repository and keeping the language model up to date throughout development. All source code, documentation-related source code, and data files will be included at the end of the iteration. The executable and any other deliverables will also be provided at the end of the iteration.

5. Miscellaneous

There are no miscellaneous items at this time.